

Worksheet 5

Time EvolutionOperators • Eigenvalues • Hamiltonian • Schrödinger • Evolving states

1. The Hamiltonian of a two-state system is given by

$$\hat{H} \rightarrow \begin{pmatrix} E_1 & 0 \\ 0 & E_2 \end{pmatrix}.$$

What basis is this Hamiltonian written in? How do you know?

2. What do the values E_1 and E_2 represent? What do we call them? Write the energy eigenstates – call them $|E_1\rangle$ and $|E_2\rangle$ – in matrix notation.

3. A second observable, which we'll call A , can be written in the energy basis as

$$\hat{A} \rightarrow \begin{pmatrix} 0 & a \\ a & 0 \end{pmatrix}.$$

Do $\hat{H}$ and $\hat{A}$ commute? What does that imply about the two observables?

4. Find the eigenvalues and eigenstates of the operator $\hat{A}$. What do the eigenvalues represent? What about the eigenstates?

5. Suppose at time $t = 0$, our quantum system is in a state given by

$$|\psi(0)\rangle = | + a \rangle.$$

If we measure A on this state, what are the possible results and the probabilities of each?

6. Write the state $|\psi(0)\rangle$ in the energy basis, both in Dirac and matrix notation. What is the exact value of the energy of the system at $t = 0$?
7. What if we go ahead and measure the energy E – what are the possible results and the probabilities of each?
8. What is the state at a later time? Write it in the energy basis.
9. Write the state using the eigenstates of A as your basis.

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